National University
of Computer & Emerging Sciences
Islamabad

MT-1003 Calculus and Analytic Geometry

Assignment No: 03 | **Individual Assignment**

Semester: Fall 2023 | **Marks:** 10x10

Due date: Monday, 1st November, 11:59pm | **Section:** BSCS all sections

Instructors: Dr Hamda, Dr Imran Shahzad and Mr. Usman Rashid

Instructions:

1. Plagiarized work will result in zero marks.
2. No retake or late submission will be accepted.
3. Attach complete code, results, and screenshot for questions that require programming solution.
4. Questions that show the icon require partial or complete solution using MATLAB.
5. Submit the assignment on google classroom before the deadline.
6. You must submit the assignment in a compiled form, i.e., a single PDF file of your complete assignment including programming and non-programming related questions.
7. The PDF file should be according to the following format: Id_section_A1 e.g., i22-123456 A2. A2 in the end denotes Assignment 2.
8. The images of by-hand solution should be properly scanned. You can use any mobile application such as Cam Scanner or Adobe Scan for scanning. Each of these applications allow you to export pdf or image files which you can use to combine with your programming solutions. Do not attach direct images from the camera application of your mobile phone, or screenshots.
9. You can use MATLAB tutorials shared on google classroom. If you do not have MATLAB, you can use Octave Online as well.

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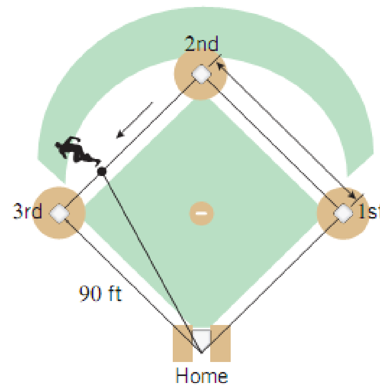
Related Rates:
Question 01



A baseball diamond is a square whose sides are 90 ft long. Suppose that a player running from second base to third base has a speed of 30 ft/s at the instant when he is 20 ft from third base. At what rate is the player's distance from home plate changing at that instant?

Question 02

An aircraft is flying horizontally at a constant height of 4000 ft above a fixed observation point. At a certain instant the angle of elevation θ is 30◦ and decreasing, and the speed of the aircraft is 300 mi/h.

(a) How fast is θ decreasing at this instant? Express the result in units of deg/s.

(b) How fast is the distance between the aircraft and the observation point changing at this instant? Express the result in units of ft/s. Use 1 mi = 5280 ft.

Linear Approximation:
Question 03
Use the linear approximation of $(1 + x)^k$ to estimate the following:

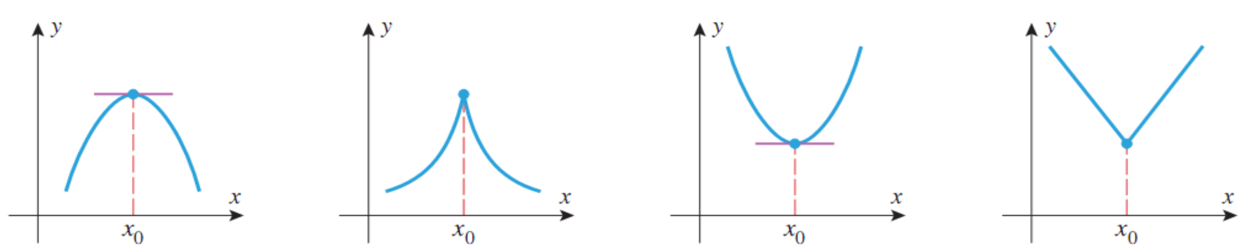
(a) $(1.002)^{10}$
(b) $(1.0002)^{10}$
(d) $(1.002)^{20}$
(e) $(1.0002)^{20}$

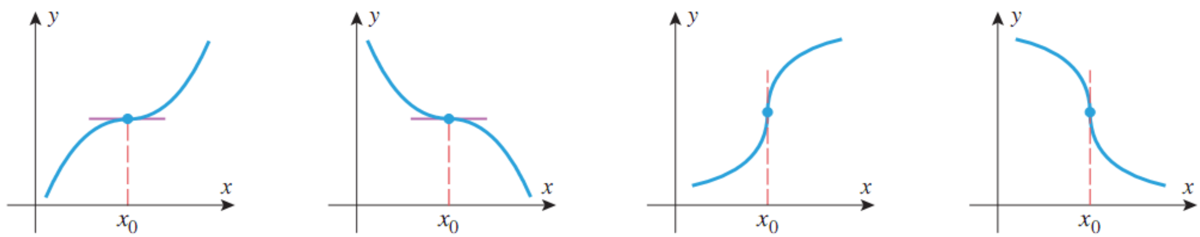
Question 04

Find the linearization of the function $f(x) = \sqrt{x + 3}$ at a=1 and use it to approximate the numbers $\sqrt{3.98}$ *and* $\sqrt{4.05}$. Are these approximations overestimates or underestimates?

Applications of Derivatives:
Question 05
For the following EIGHT graphs, explain whether there is

(a) A relative maxima/ minima at x_o
(b) Stationary point at x_o
(b) Critical point at x_o
(c) Inflection point at x_o





Question 06

Let $f(x) = 3x^5 - 5x^3$ be a function.

Without evaluating the derivatives, give your opinion about

(a) Number of critical points
(b) If x_o is a critical point then what is another critical point.
(c) If m is a local minimum at x_o then what about its local maxima? Whether it exists or not.
(d) Is there any point of inflection? If yes, is there any relation of it with the two other critical points? Explain.

Question 07

Find the relative extrema $f(x) = |sin2x|$ in the interval $0 < x < 2\pi$, and confirm that your results are consistent with the graph of $f(x)$ generated with **MATLAB**.

Question 08

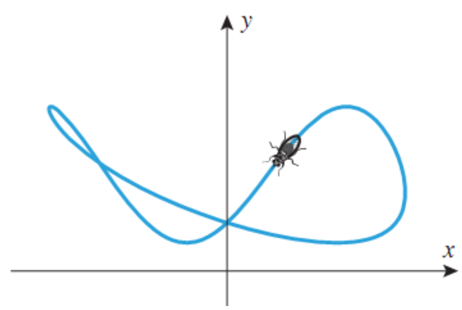
Use a **MATLAB** to generate the graph of $f(x) = 6x^{\frac{1}{3}} + 3x^{\frac{3}{4}}$, and discuss what it tells you about relative extrema, inflection points, asymptotes, and end behavior. Use calculus to find the exact locations of all key features of the graph.

Question 09
The accompanying figure shows the path of a fly whose equations of motion are

$x = \frac{cost}{2+sint}$ and $y = 3 + sin2t - 2sin^2t \quad (0 \le t \le 2\pi)$

(a) How high and low does it fly?

(b) How far left and right of the origin does it fly?



Applied Optimization:
Question 10

A farmer has 2400 ft of fencing and wants to fence off a rectangular field that borders one side by a straight river. He needs no fence along the river. What are the dimensions of the field that has the largest area?